

CS 4300/6300: Artificial Intelligence

Probability

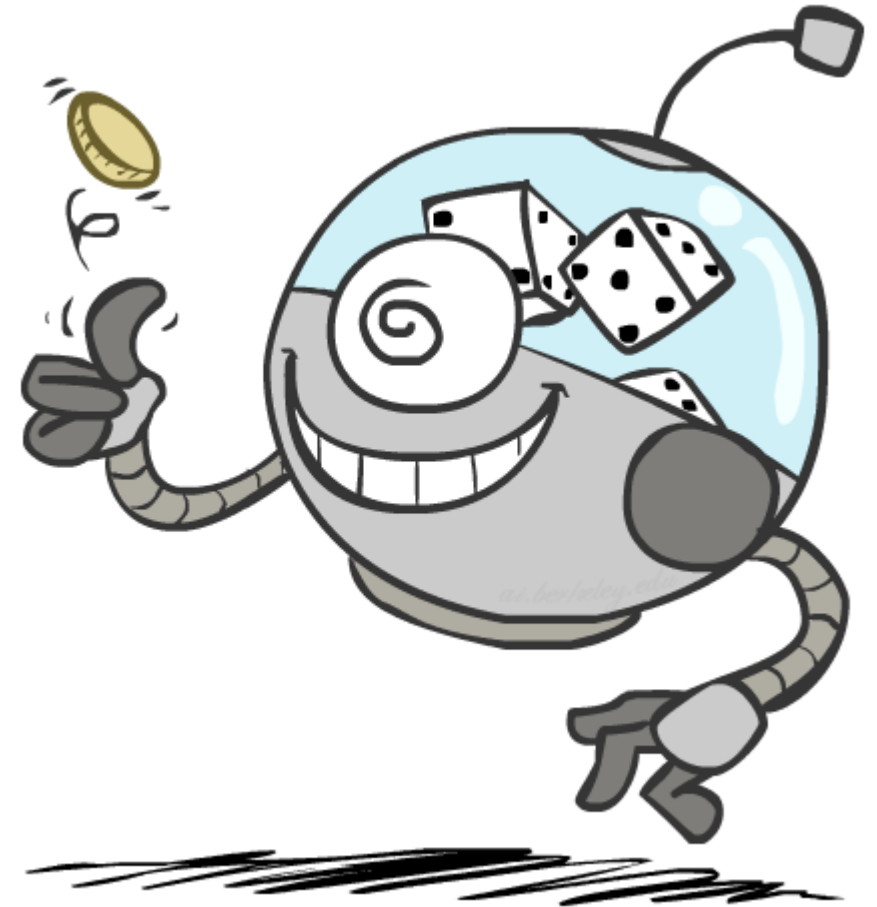


Instructor: Daniel Brown --- University of Utah

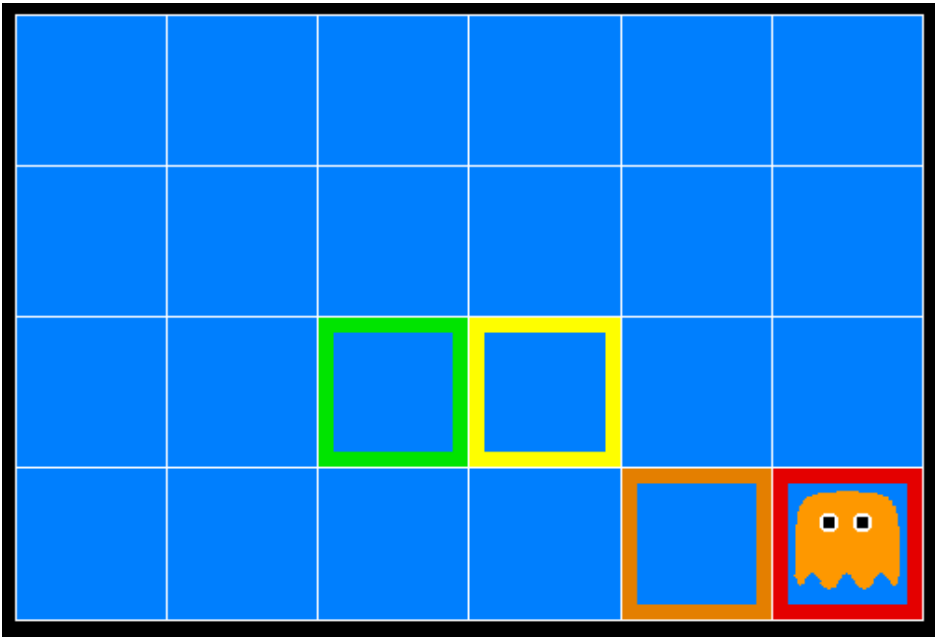
[Based on slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. <http://ai.berkeley.edu>.]

Today

- Probability
 - Random Variables
 - Joint and Marginal Distributions
 - Conditional Distribution
 - Product Rule, Chain Rule, Bayes' Rule
 - Inference
 - Independence
- You'll need all this stuff A LOT for the next few weeks, so make sure you go over it now!



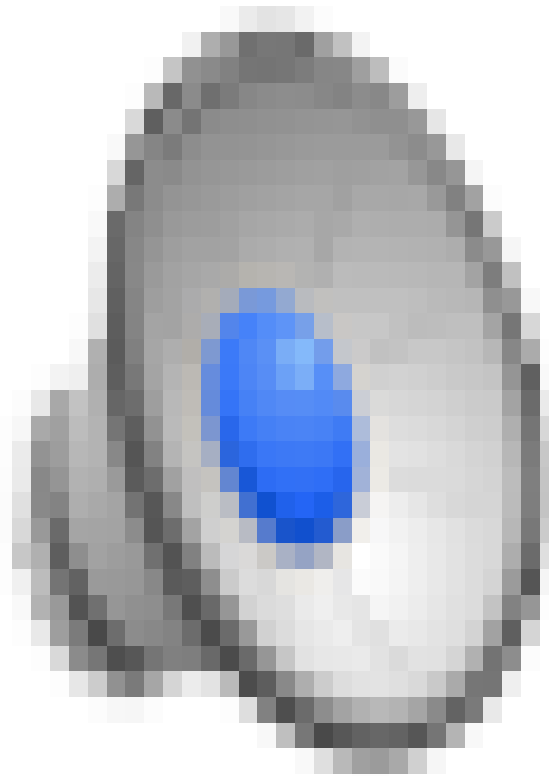
- Sensors are noisy, but we know $P(\text{Color} \mid \text{Distance})$



P(red 3)	P(orange 3)	P(yellow 3)	P(green 3)
0.05	0.15	0.5	0.3

[Demo: Ghostbuster – no probability (L12D1)]

Video of Demo Ghostbuster – No probability



Uncertainty

- General situation:
 - **Observed variables (evidence):** Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)
 - **Unobserved variables:** Agent needs to reason about other aspects (e.g. where an object is or what disease is present)
 - **Model:** Agent knows something about how the known variables relate to the unknown variables
- Probabilistic reasoning gives us a framework for managing our beliefs and knowledge

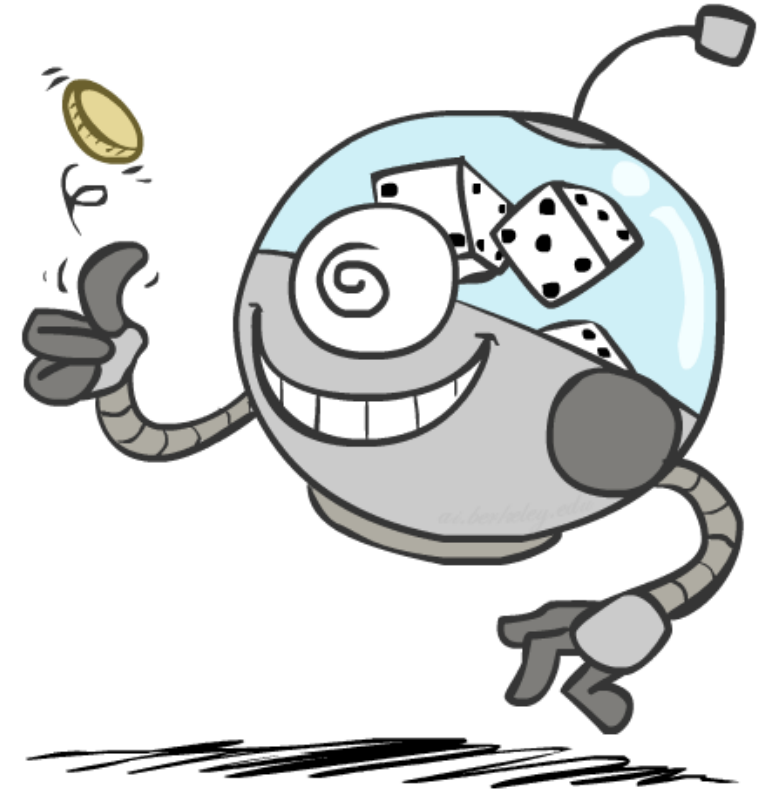
0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

0.17	0.10	0.10
0.09	0.17	0.10
<0.01	0.09	0.17

<0.01	<0.01	0.03
<0.01	0.05	0.05
<0.01	0.05	0.81

Random Variables

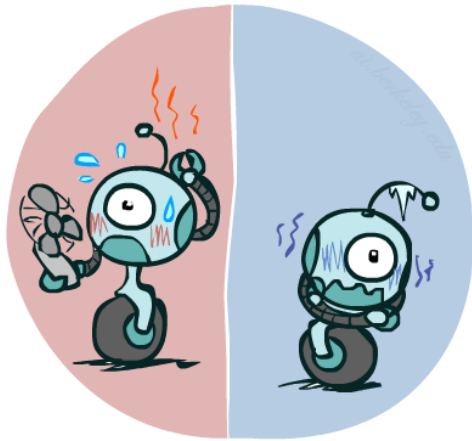
- A random variable is some aspect of the world about which we (may) have uncertainty
 - R = Is it raining?
 - T = Is it hot or cold?
 - D = How long will it take to drive to work?
 - L = Where is the ghost?
- We denote random variables with capital letters



Probability Distributions

- Associate a probability with each value

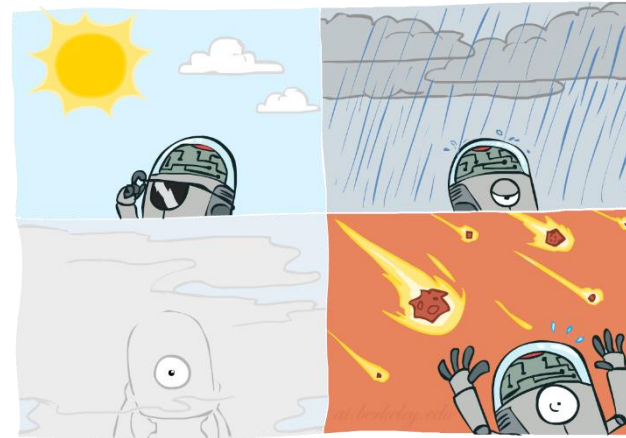
- Temperature:



$P(T)$

T	P
hot	0.5
cold	0.5

- Weather:



$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

$10e^{-8}$

Probability Distributions

- Unobserved random variables have distributions

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Shorthand notation:

$$P(\textit{hot}) = P(T = \textit{hot}),$$

$$P(\textit{cold}) = P(T = \textit{cold}),$$

$$P(\textit{rain}) = P(W = \textit{rain}),$$

...

OK if all domain entries are unique

- A distribution is a TABLE of probabilities of values
- A probability (lower case value) is a single number

$$P(W = \textit{rain}) = 0.1$$

- Must have: $\forall x \ P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A *joint distribution* over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey: $P(x_1, x_2, \dots, x_n) \geq 0$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Size of distribution if n variables with domain sizes d ?

d^n

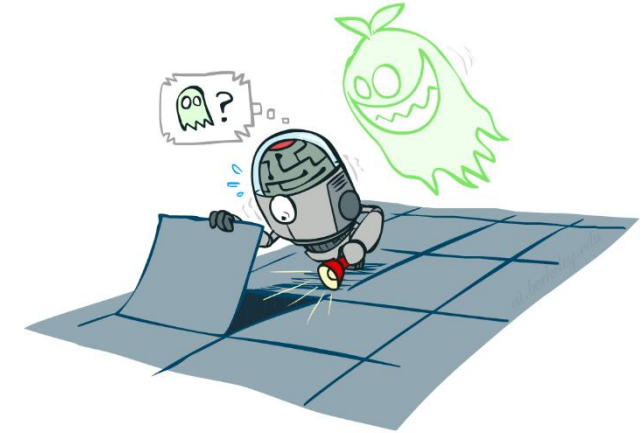
- For all but the smallest distributions, impractical to write out!

Probabilistic Models

- A probabilistic model is a joint distribution over a set of random variables
- Probabilistic models:
 - (Random) variables with domains
 - Assignments are called *outcomes*
 - Joint distributions: say whether assignments (outcomes) are likely
 - *Normalized*: sum to 1.0
 - Ideally: only certain variables directly interact

Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



Practice: Events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- From a joint distribution, we can calculate the probability of any event

- Probability that it's hot AND sunny?

0.4

- Probability that it's hot?

0.5

- Probability that it's hot OR sunny?

0.7

- Typically, the events we care about are *partial assignments*, like $P(T=\text{hot})$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Events

■ $P(+x, +y) ?$ 0.2

■ $P(+x) ?$ 0.5

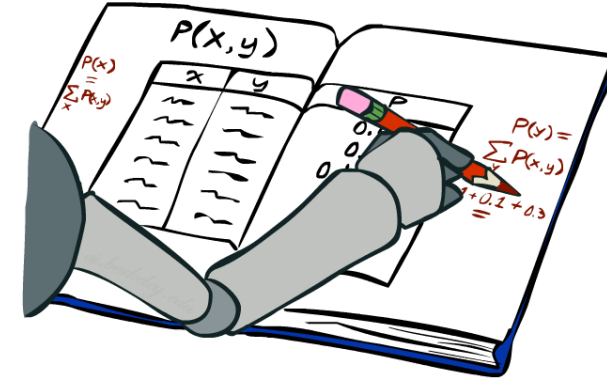
■ $P(-y \text{ OR } +x) ?$ 0.7

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\sum_w P(t, w)$$



$$P(t) = \sum_w P(t, w)$$

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4



$$P(w) = \sum_t P(t, w)$$

$$\sum_t P(t, \text{rain})$$

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Practice: Marginal Distributions

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1



$$P(x) = \sum_y P(x, y)$$



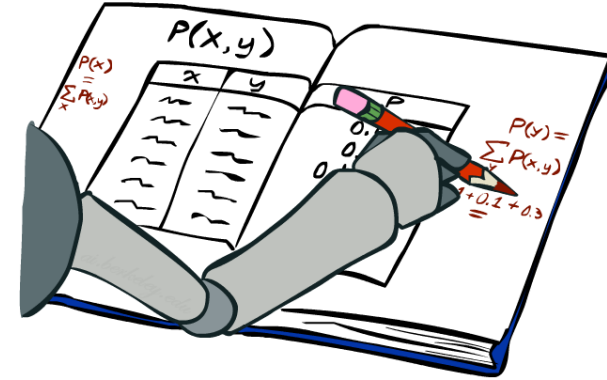
$$P(y) = \sum_x P(x, y)$$

$P(X)$

X	P
+x	0.5
-x	0.5

$P(Y)$

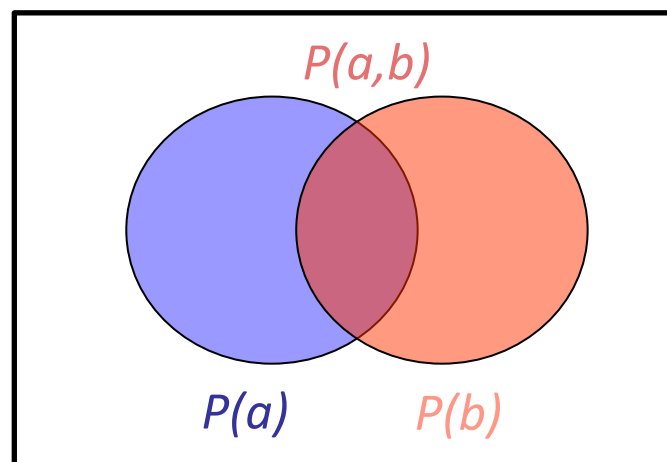
Y	P
+y	0.6
-y	0.4



Conditional Probabilities

- A simple relation between joint and conditional probabilities
 - In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a, b)}{P(b)}$$



$P(T)$

h	0.5
c	0.5

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s | T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned}
 &= P(W = s, T = c) + P(W = r, T = c) \\
 &= 0.2 + 0.3 = 0.5
 \end{aligned}$$

Canvas Quiz: Conditional Probabilities

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

- $P(+x \mid +y) ? = \frac{P(+x, +y)}{P(+y)} = \frac{0.2}{0.6} = \frac{1}{3}$

- $P(-x \mid +y) ? = 1 - \frac{1}{3} = \frac{2}{3}$

- $P(-y \mid +x) ? = \frac{P(-y, +x)}{P(+x)} = \frac{0.3}{0.5} = \frac{3}{5}$

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W T)$	$P(W T = hot)$	
	W	P
	sun	0.8
	rain	0.2
	$P(W T = cold)$	
	W	P
	sun	0.4
	rain	0.6

Joint Distribution

$P(T, W)$		
T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Normalization Trick

$$\sum_{w \in W} P(W=w, T=c)$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\begin{aligned} P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\ &= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\ &= \frac{0.2}{0.2 + 0.3} = 0.4 \end{aligned}$$



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

$$\begin{aligned} P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\ &= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\ &= \frac{0.3}{0.2 + 0.3} = 0.6 \end{aligned}$$

Normalization Trick

$$\begin{aligned}
 P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\
 &= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\
 &= \frac{0.2}{0.2 + 0.3} = 0.4
 \end{aligned}$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

$$\begin{aligned}
 P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\
 &= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\
 &= \frac{0.3}{0.2 + 0.3} = 0.6
 \end{aligned}$$

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

- Why does this work? Sum of selection is $P(\text{evidence})$! ($P(T=c)$, here)

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

Practice: Normalization Trick

■ $P(X \mid Y=-y)$?

$P(X, Y)$

X	Y	P
+ x	+ y	0.2
+x	-y	0.3
- x	+ y	0.4
-x	-y	0.1

SELECT the joint probabilities matching the evidence



$P(X \mid Y=-y)$

+x	-y	0.3
-x	-y	0.1

0.4

NORMALIZE the selection
(make it sum to one)



$P(X \mid Y=-y)$

+x	3/4
-x	1/4

To Normalize

- (Dictionary) To bring or restore to a normal condition

All entries sum to ONE

- Procedure:

- Step 1: Compute $Z = \text{sum over all entries}$
- Step 2: Divide every entry by Z

- Example 1

W	P
sun	0.2
rain	0.3

Normalize
→
 $Z = 0.5$

W	P
sun	0.4
rain	0.6

- Example 2

T	W	P
hot	sun	20
hot	rain	5
cold	sun	10
cold	rain	15

Normalize
→
 $Z = 50$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Inference

- Probabilistic inference: compute a desired probability from other known probabilities (e.g. conditional from joint)
- We generally compute conditional probabilities
 - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
 - These represent the agent's *beliefs* given the evidence
- Probabilities change with new evidence:
 - $P(\text{on time} \mid \text{no accidents, 5 a.m.}) = 0.95$
 - $P(\text{on time} \mid \text{no accidents, 5 a.m., raining}) = 0.80$
 - Observing new evidence causes *beliefs to be updated*



Inference by Enumeration

- General case:

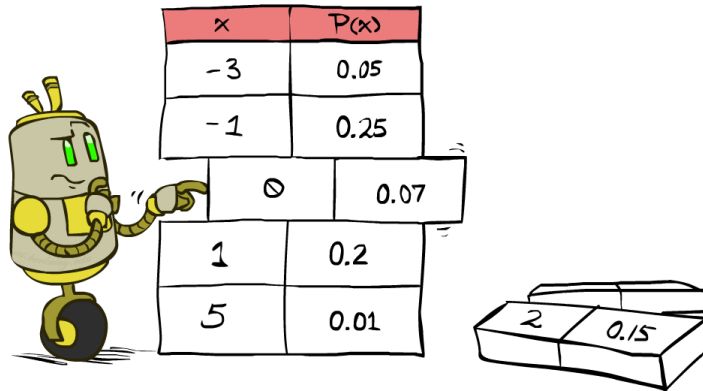
- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $$\left. \begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array} \right\} \begin{array}{l} X_1, X_2, \dots X_n \\ \text{All variables} \end{array}$$

- We want:

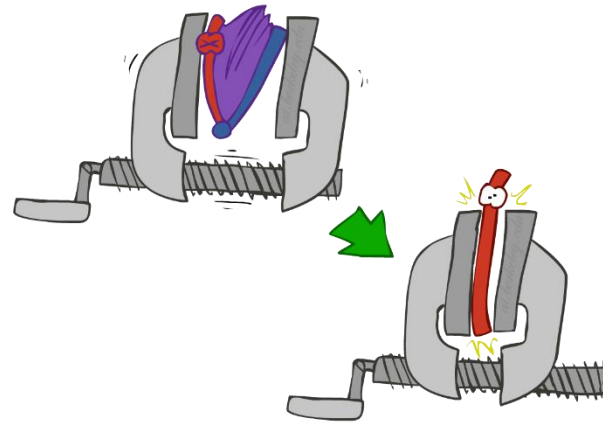
** Works fine with multiple query variables, too*

$$P(Q|e_1 \dots e_k)$$

- Step 1: Select the entries consistent with the evidence



- Step 2: Sum out H to get joint of Query and evidence



- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots X_n}$$

Inference by Enumeration

- $P(W)$?

W	P(W)
S	0.65
r	0.35

- $P(W \mid \text{winter})$?

W	P(W) S=winter
S	0.5
r	0.5

- $P(W \mid \text{winter, hot})$?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

2/3
1/3

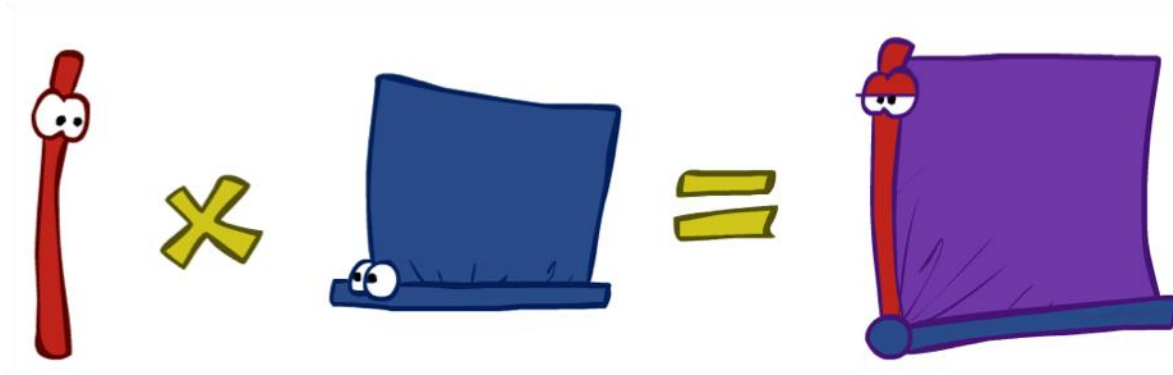
Inference by Enumeration

- Obvious problems:
 - Worst-case time complexity $O(d^n)$
 - Space complexity $O(d^n)$ to store the joint distribution

The Product Rule

- Sometimes we have conditional distributions but want the joint distribution:

$$P(y)P(x|y) = P(x, y) \quad \longleftrightarrow \quad P(x|y) = \frac{P(x, y)}{P(y)}$$



The Product Rule

$$P(y)P(x|y) = P(x, y)$$

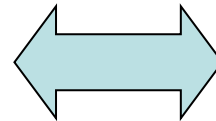
- Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$P(D, W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

The Chain Rule

- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

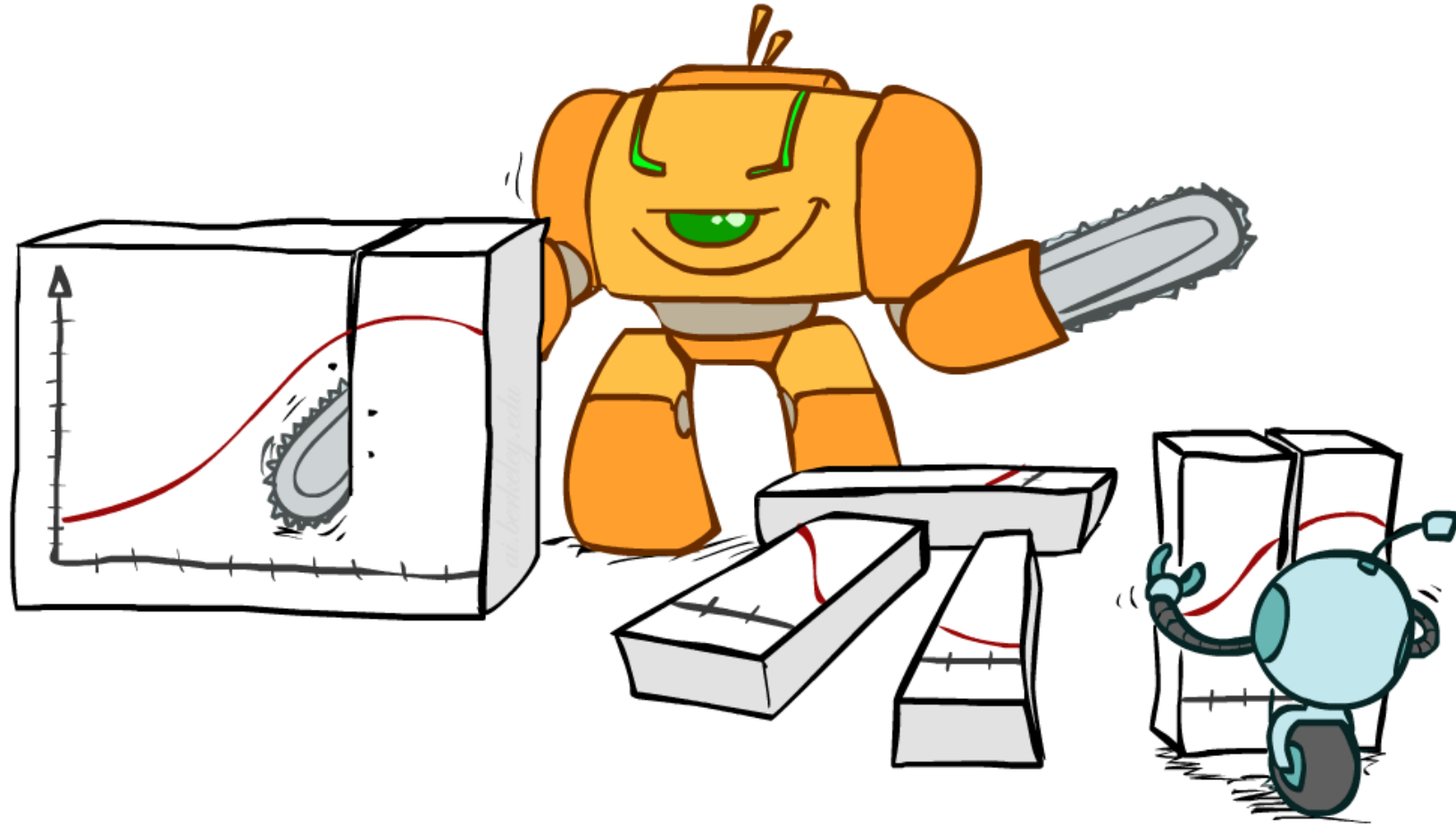
Handwritten red annotations showing the chain rule for three variables:

$$P(x_1, x_2, x_3) = \frac{P(x_1, x_2, x_3)}{P(x_1, x_2)} \cdot \frac{P(x_1, x_2)}{P(x_1)}$$

The terms $P(x_1)$ and $P(x_1, x_2)$ in the denominators are crossed out with red lines.

- You can pick any order.
- Why is the Chain Rule always true?

Bayes Rule



Bayes' Rule

- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

That's my rule!

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

$P(\text{cough}|\text{covid})$
Know
 $P(\text{covid}|\text{cough})$

- Why is this at all helpful?

- Lets us build one conditional from its reverse
- Often one conditional is tricky but the other one is simple
- Foundation of many systems (e.g. ASR, MT, IRL)

- In the running for most important AI equation!



Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

$$\overset{\text{posterior}}{P(\text{cause}|\text{effect})} = \frac{\overset{\text{likelihood}}{P(\text{effect}|\text{cause})} \overset{\text{prior}}{P(\text{cause})}}{P(\text{effect})} = \sum_c P(e, c) = \sum_c P(e|c) P(c)$$

- Example:

- M: meningitis, S: stiff neck

$$\left. \begin{aligned} P(+m) &= 0.0001 \\ P(+s|+m) &= 0.8 \\ P(+s|-m) &= 0.01 \end{aligned} \right\} \text{Example gives}$$

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.999}$$

About 0.008

- Note: posterior probability of meningitis still very small
 - Note: you should still get stiff necks checked out! Why?

Quiz: Bayes' Rule

- Given:

$$P(W)$$

R	P
sun	0.8
rain	0.2

$$P(D|W)$$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

- What is $P(W \mid \text{dry})$?

$$P(\text{sun}|\text{dry}) = \frac{P(\text{dry}|\text{sun})P(\text{sun})}{P(\text{dry})} = \frac{P(\text{dry}|\text{sun})P(\text{sun})}{P(\text{dry}|\text{sun})P(\text{sun}) + P(\text{dry}|\text{rain})P(\text{rain})} = \frac{0.9 * 0.8}{0.9 * 0.8 + 0.3 * 0.2} \approx 0.92$$

$$P(\text{rain}|\text{dry}) = \frac{P(\text{dry}|\text{rain})P(\text{rain})}{P(\text{dry})} = \frac{P(\text{dry}|\text{rain})P(\text{rain})}{P(\text{dry}|\text{sun})P(\text{sun}) + P(\text{dry}|\text{rain})P(\text{rain})} = \frac{0.3 * 0.2}{0.9 * 0.8 + 0.3 * 0.2} \approx 0.08$$

Ghostbusters, Revisited

- Let's say we have two distributions:
 - Prior distribution** over ghost location: $P(G)$
 - Let's say this is uniform
 - Sensor reading model: $P(R | G)$
 - Given: we know what our sensors do
 - R = reading color measured at $(1,1)$
 - E.g. $P(R = \text{yellow} | G=(1,1)) = 0.1$
- We can calculate the **posterior distribution** $P(G|r)$ over ghost locations given a reading using Bayes' rule:

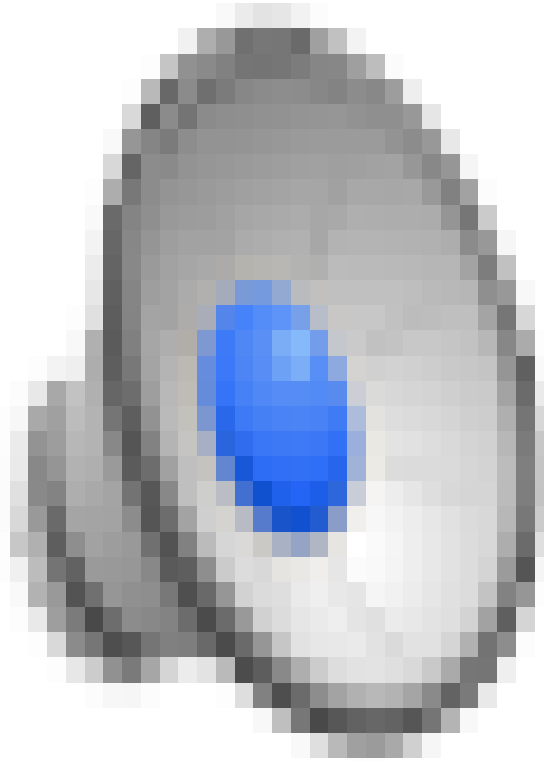
$$P(g|r) \propto P(r|g)P(g)$$



0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

0.17	0.10	0.10
0.09	0.17	0.10
<0.01	0.09	0.17

Video of Demo Ghostbusters with Probability



Independence

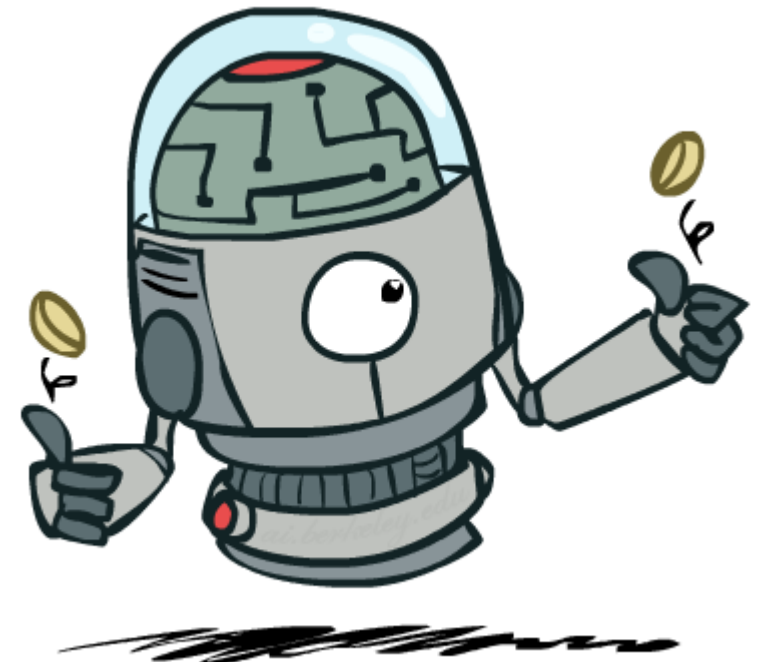
- Two variables are *independent* in a joint distribution if:

$$P(X, Y) = P(X)P(Y)$$

$$X \perp\!\!\!\perp Y$$

$$\forall x, y \ P(x, y) = P(x)P(y)$$

- Says the joint distribution *factors* into a product of two simple ones
 - Usually variables aren't independent!
- Can use independence as a *modeling assumption*
 - Independence can be a simplifying assumption
 - Empirical* joint distributions: at best “close” to independent
 - What could we assume for {Weather, Traffic, Cavity}?



Example: Independence?

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

$$P_2(T, W) = P(T)P(W)$$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

Example: Independence

- N fair, independent coin flips:

$P(X_1)$

H	0.5
T	0.5

$P(X_2)$

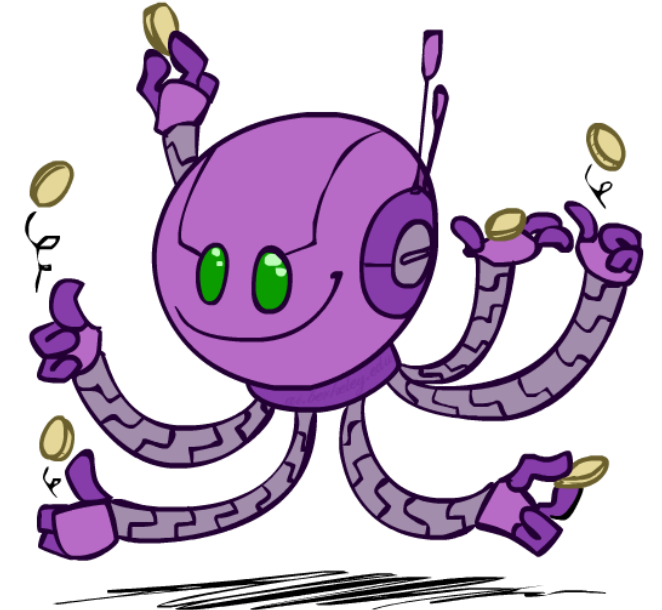
H	0.5
T	0.5

...

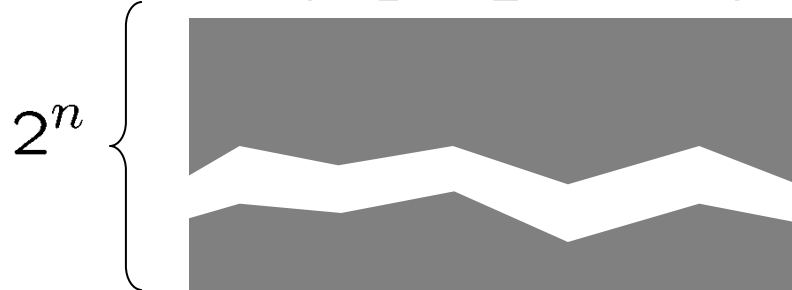
$2 \times n$

$P(X_n)$

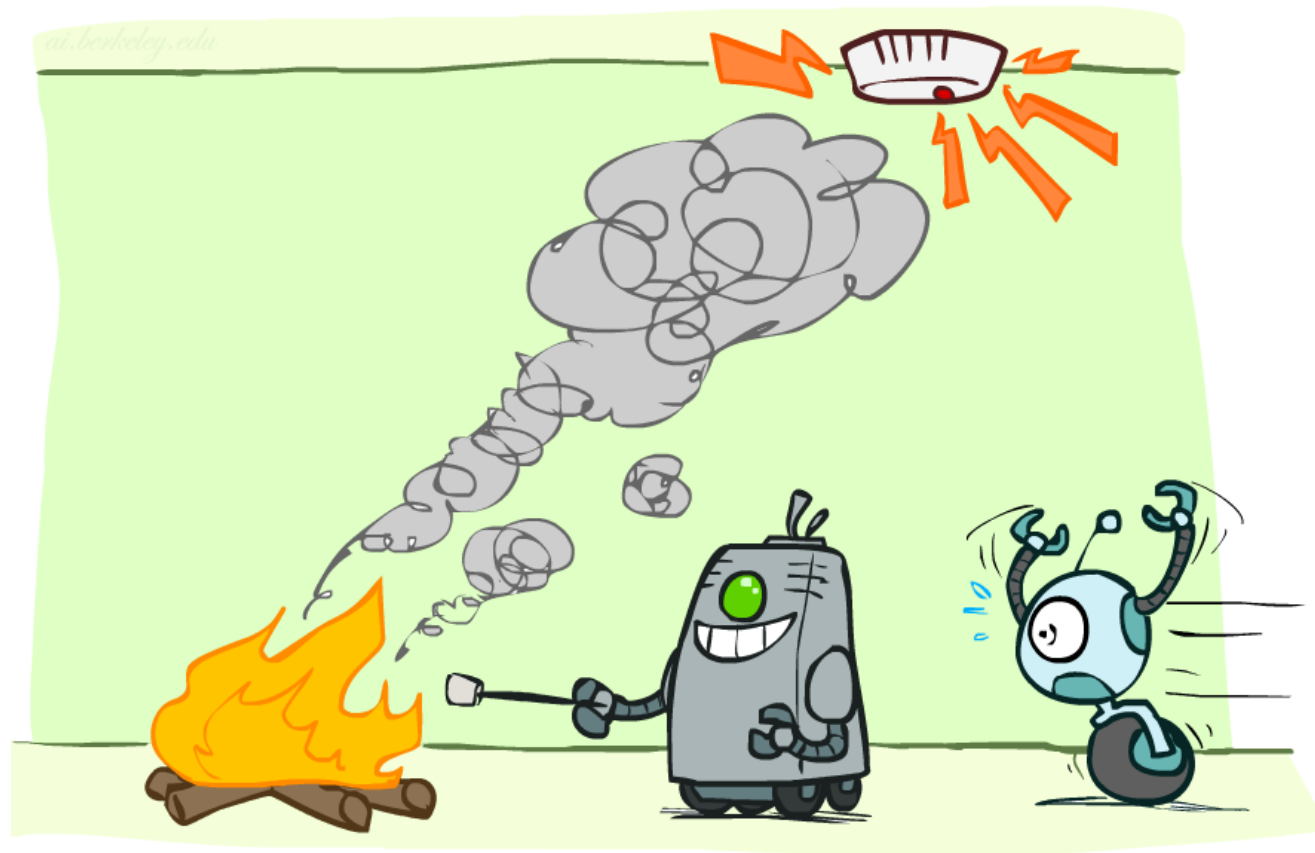
H	0.5
T	0.5



$P(X_1, X_2, \dots, X_n)$

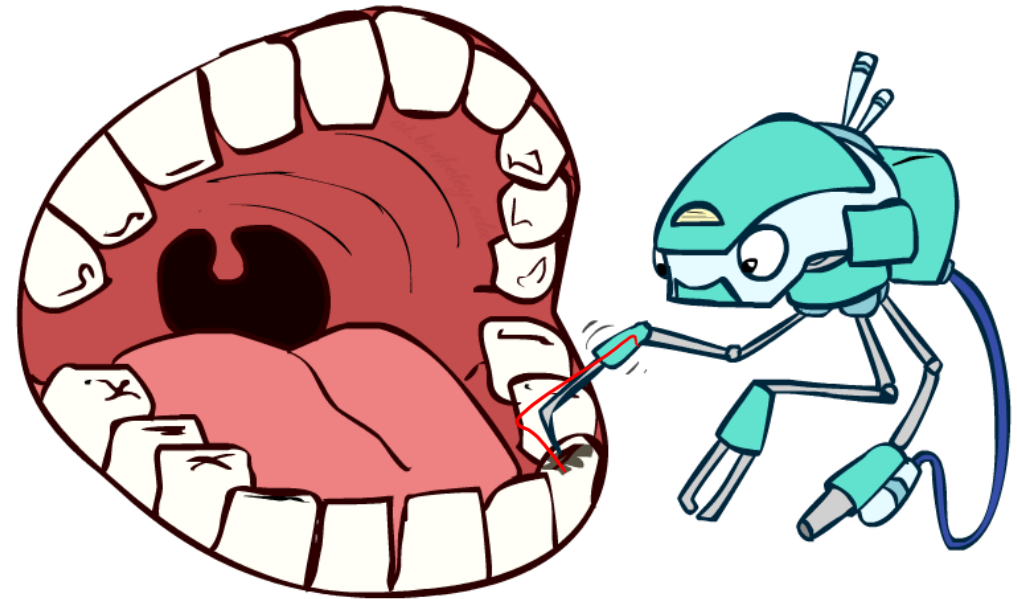


Conditional Independence



Conditional Independence

- $P(\text{Toothache}, \text{Cavity}, \text{Catch})$
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
 - $P(+\text{catch} \mid +\text{toothache}, +\text{cavity}) = P(+\text{catch} \mid +\text{cavity})$
- The same independence holds if I don't have a cavity:
 - $P(+\text{catch} \mid +\text{toothache}, -\text{cavity}) = P(+\text{catch} \mid -\text{cavity})$
- Catch is *conditionally independent* of Toothache given Cavity:
 - $P(\text{Catch} \mid \text{Toothache}, \text{Cavity}) = P(\text{Catch} \mid \text{Cavity})$
- Equivalent statements:
 - $P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$
 - $P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$
 - One can be derived from the other easily



Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.

- X is conditionally independent of Y given Z

$$X \perp\!\!\!\perp Y | Z$$

if and only if:

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

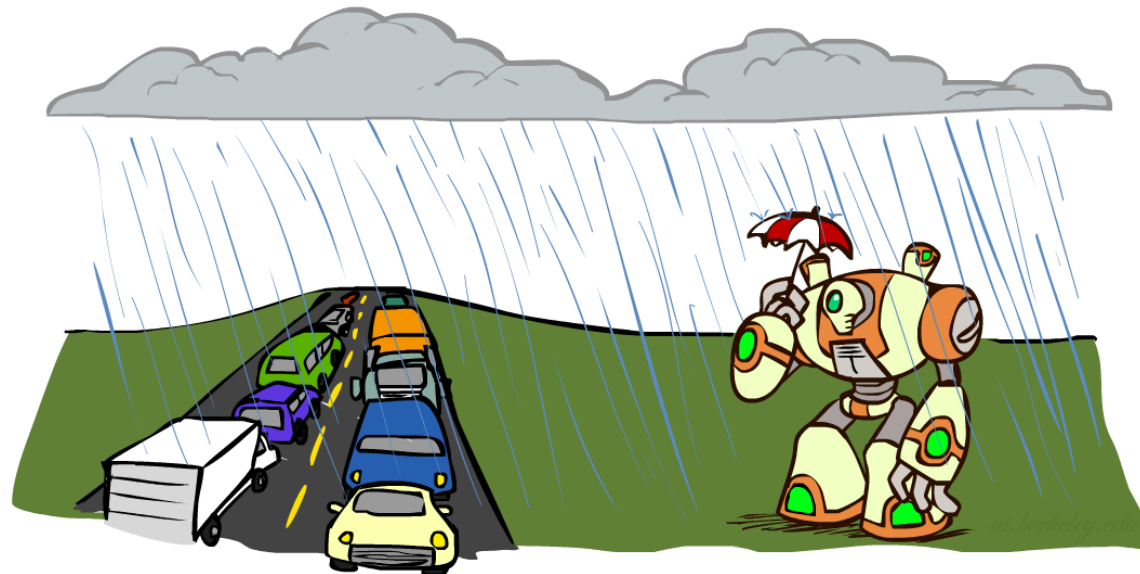
or, equivalently, if and only if

$$\forall x, y, z : P(x | z, y) = P(x | z)$$

Conditional Independence

- What about this domain:

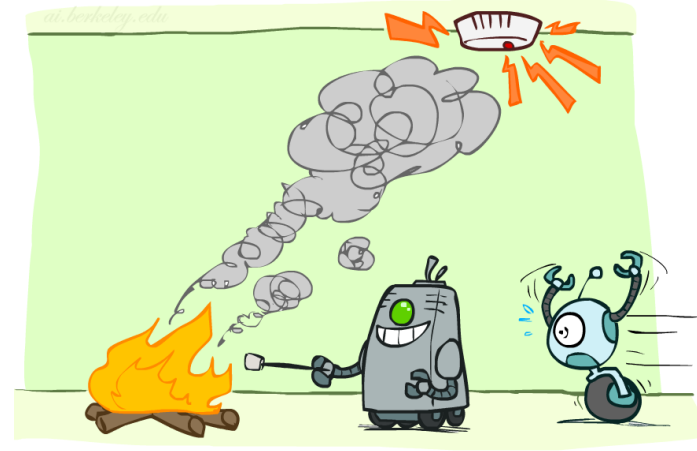
- Traffic
- Umbrella
- Raining



Conditional Independence

- What about this domain:

- Fire
- Smoke
- Alarm



Probability Recap

- Conditional probability $P(x|y) = \frac{P(x, y)}{P(y)}$
- Product rule $P(x, y) = P(x|y)P(y)$
- Chain rule
$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$
- X, Y independent if and only if: $\forall x, y : P(x, y) = P(x)P(y)$
- X and Y are conditionally independent given Z if and only if: $X \perp\!\!\!\perp Y | Z$
 $\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$

Bayes' Rule

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Next Time: MDPs
